Lighthouse

Building an agent-guided mobile robot from an empty Gazebo world

SLAM · Nav2 · frontier exploration · natural-language control · preemption

Dan · 2026-10-26

THE GOAL

One sentence, and everything else serves it

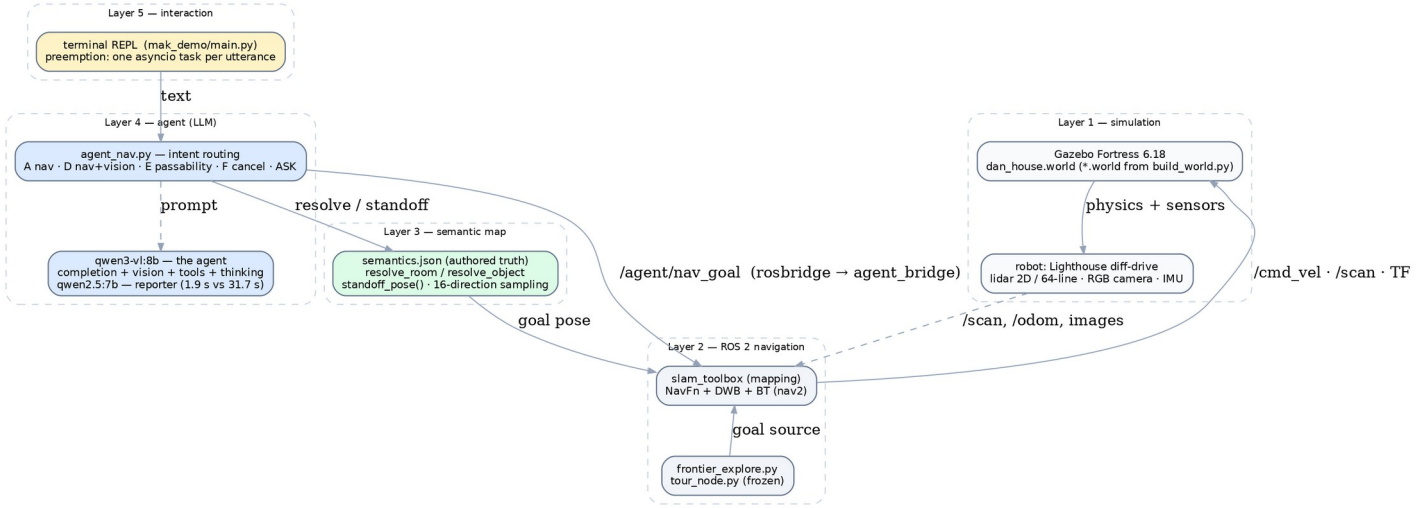
「带我去厨房看看冰箱」 — "take me to the kitchen and look at the fridge"

- Resolve the words to a place and an object
- Drive there without hitting furniture
- Stop at a pose from which the camera can actually see the object
- Answer what it saw — or refuse honestly if the place does not exist
- Abandon the trip cleanly if the operator changes their mind

Design driver: every failure must be diagnosable, not merely observed.

ARCHITECTURE

Five layers, one authority rule



- Simulation owns physics and sensors
- Nav2 owns motion, never intent
- The semantic map owns naming (a database)
- The agent owns intent; code decides the route
- Interaction owns concurrency (one task per utterance)

Fig. Decisions flow down as goals; measurements flow up as observations. The model never touches /cmd_vel.

A generated house, not a hand-drawn one

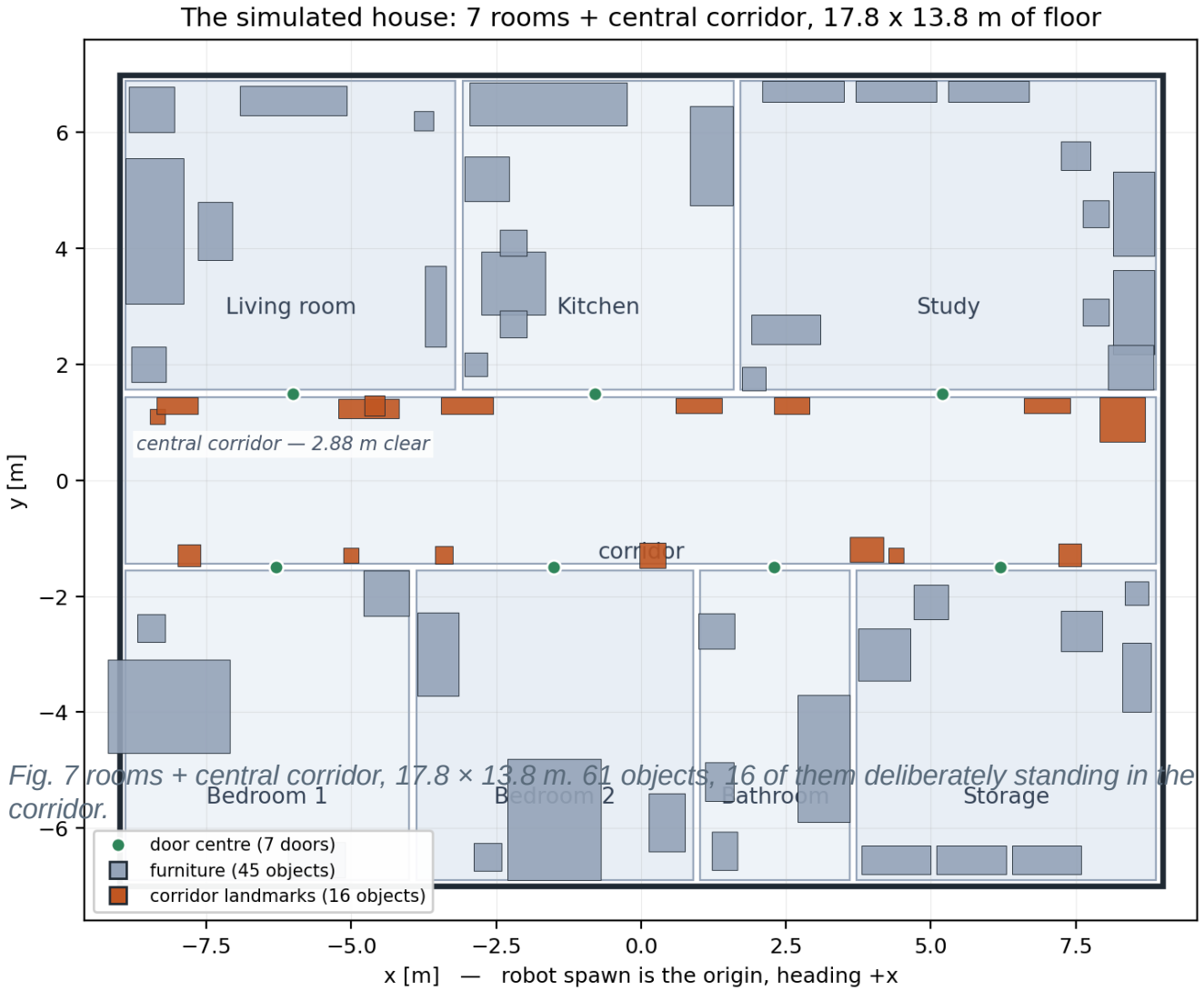


Fig. 7 rooms + central corridor, 17.8 × 13.8 m. 61 objects, 16 of them deliberately standing in the corridor.

- build_world.py (898 lines) is the only source of world truth
- It also generates semantics.json and the 32-waypoint route
- 2D world raises low furniture so a single-line lidar can see it
- 3D world keeps real heights
- Hand-editing a .world file is forbidden by rule

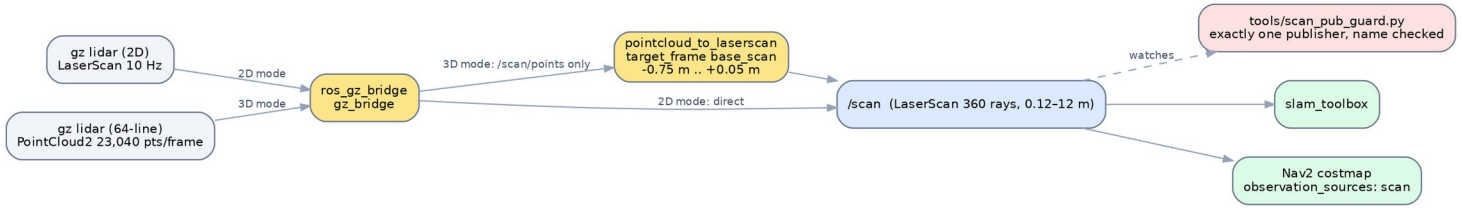
SENSORS

Two lidar heads, one camera without depth

Sensor	Rate	Good for	Must not be used for
/scan (LaserScan)	8.4–8.5 Hz	SLAM, costmaps, passability, frontiers	seeing above the slice
/scan/points (64-line)	8.4 Hz	future 3D costmap, offline analysis	anything today — nothing subscribes
/camera/image_raw	7.1 Hz, 320×240	VLM identity check, passability veto	geometry, scale, depth
/imu	100 Hz	EKF experiments	accepted navigation
/odom	50 Hz	truth-pose navigation, TF	real-robot localisation

The camera is a semantic instrument. It has no depth channel at all.

One message type is the spine of the stack



- 360 rays, 0.12–12.0 m, frame base_scan
- A guard asserts exactly one publisher AND the right node name
- ≥ 2 publishers fails immediately: never transient
- A rogue publisher started later is caught within 30 s
- Cost of the contract: the costmap cannot see overhangs

Fig. In 3D mode the gz LaserScan is never bridged — only the point cloud, projected to a single 2D slice.

SLAM

The map everything else trusts

Property	Value	How it is known
size	359 × 278 cells @ 5 cm	PGM header
extent	17.95 × 13.90 m	cells × resolution
tilt	−0.1°	fitted in the single map decoder
interior unknown	13.5 m²	inside the house footprint only
live map (3D)	dan_house_lighthouse3d.yaml → _explore4.pgm	config/maps
localisation	loc:=truth for every acceptance run	the honest gap: loc:=amcl unverified

One decoder (tools/map_quality.py) defines free / occupied / unknown. A second decoder once reported "done" with 180 m² never seen.

The door was innocent; the goal point was guilty

- A 1.30 m door with inflation 0.45 m leaves 0.02 m per side — feasible
- Widest-path probe: every door is 0.50–0.55 m half-width → cost 0

Failed goals measured 0.35–0.40 m clearance — inside the inflation band

- Inside that band the cost is 132–177 per cell; DWB must cross all of it
- It often prefers to rotate in place instead of entering

Stall diagnosis: 419/419 trajectories valid, vx=0.000 scored 63.40 vs 72.70

→ The robot was not stuck — standing still was simply cheaper

"No valid trajectories = 0" is not a health metric.

MAPPING

Two ways to build a map, kept for two reasons

	Tour (frozen)	Frontier exploration
Driver	tour_node.py, 521 lines, plain Twist	frontier_explore.py, 943 lines
Input	32 waypoints from route.json	live SLAM map, no input
Time	~7 min of simulation	20–45 min, budget-capped
Best result	359 × 278 map, −0.1° tilt	15.8 m² interior unknown
Role	phase-1 acceptance artefact	the experiment platform

The exploration node is valuable for its cancellation machinery, not its frontier scoring.

METHOD

One run per arm is not an experiment

- 15 goals: SmacPlanner2D wins by 21 m²

40 goals, same code: NavFn wins by 18 m²

Short runs measure variance, not effect

- ≥ 3 repetitions, ≥ 40 goals, compare medians
- Always print the within-arm range next to the between-arm shift
- Isolation: fresh DDS domain + Gazebo partition + map prefix per run
- Budget by the knee of the curve (~48 goals), not by round numbers

A driver that cannot invalidate its own runs will happily produce confident nonsense.

RESULTS

More budget does not buy more map

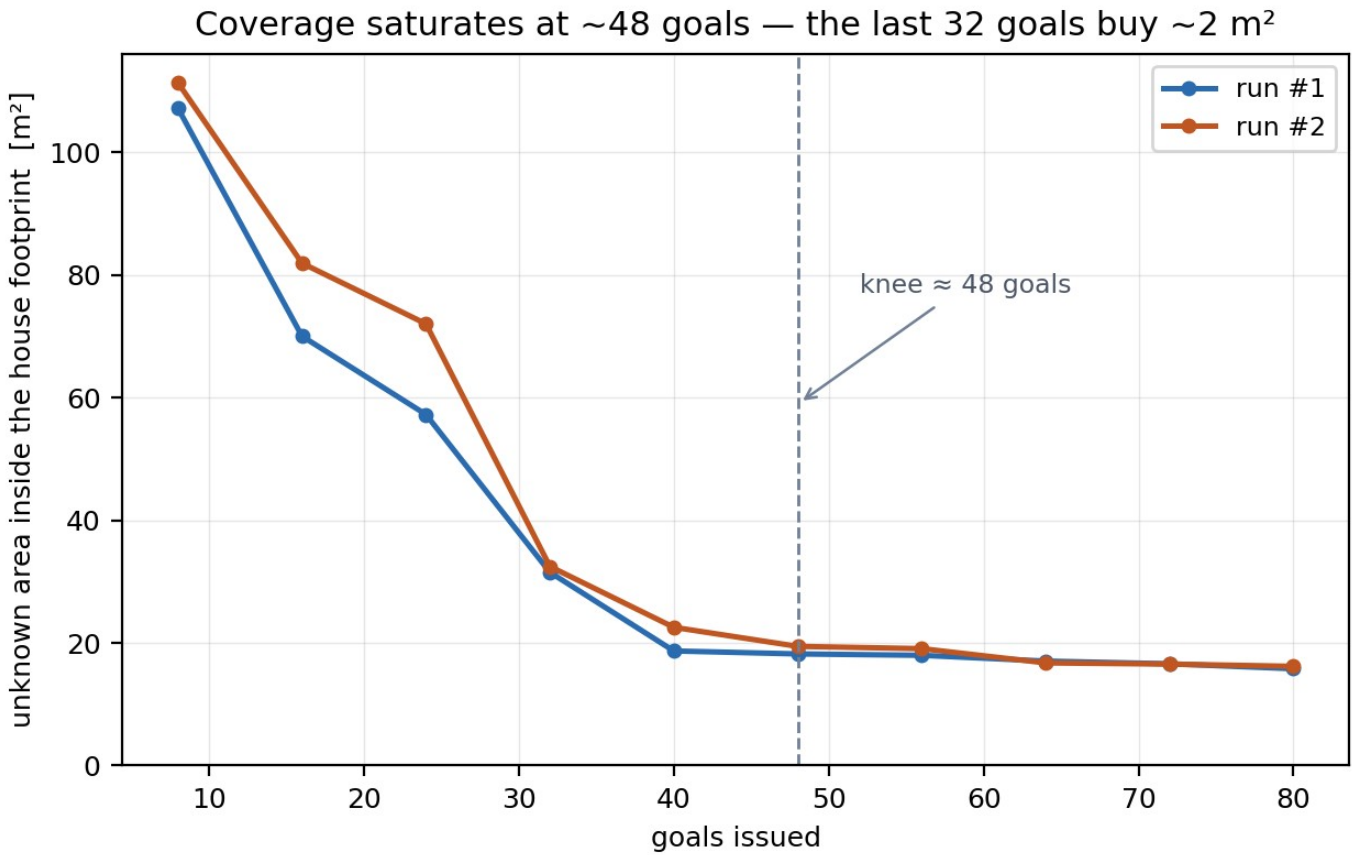
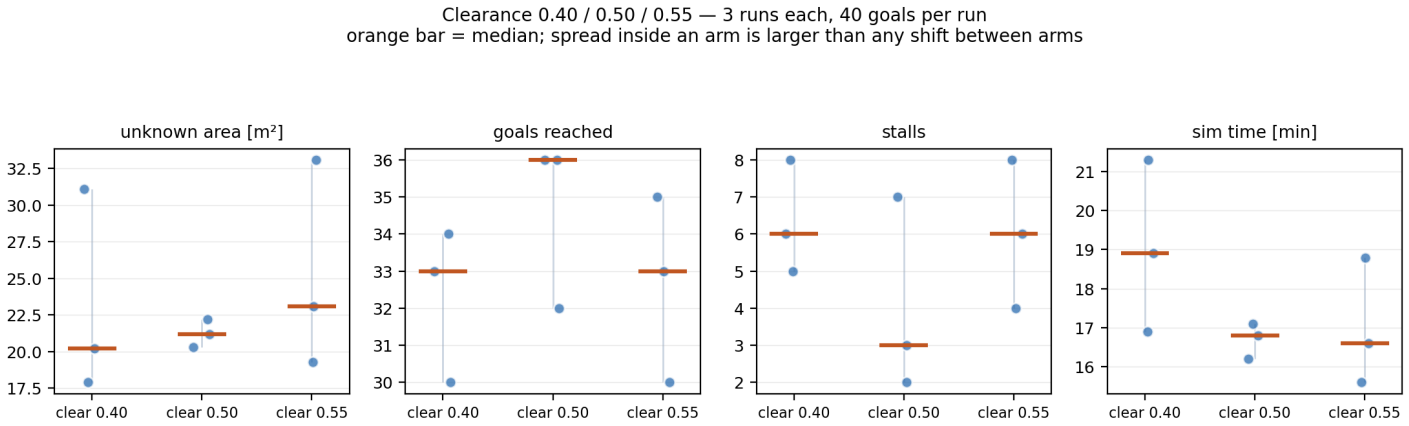


Fig. Recomputed from the checkpoint maps of both 80-goal runs. Knee at ~48 goals; the last 32 goals buy 2.2–2.9 m² in 41 minutes of simulation.

- 80 goals: 63/80 and 57/80 reached, both capped by goal count
- Stalls triple towards the cap
- Residue is shadow pockets behind furniture
- No configuration beat the 13.5 m² test map
- Recorded as a ceiling, not a target

RESULTS

A parameter that turned out not to matter



- 0.40 / 0.50 / 0.55 clearance
- Reached 33 / 36 / 33, stalls 6 / 3 / 6
- Every between-arm difference < the noise inside an arm
- Verdict: clearance is a safety floor, not a coverage knob
- One invalid run was retried on a new domain — and recorded

Fig. 3 runs per arm, 40 goals each. Between-arm median shift 2.9 m² vs within-arm range 13.8 m².

Language in, motion out — with authority kept apart

- Four processes: launch · rosbridge · agent_bridge · venv agent

Iron rule: anything importing rclpy runs on the system interpreter

- Status topic is latched, or a late subscriber waits forever
- Every goal carries a sequence number; stale results are dropped
- Agent model qwen3-vl:8b · reporter model qwen2.5:7b (1.9 s vs 31.7 s)

→ Seven tools: list_rooms, find_place, where_am_i, go_to, go_to_room, look_at, check_path

Asked to 「带我去沙发看看」, the model called find_place, answered, and never moved. tool_choice="required" was ignored by the compat layer. Hence: code moves the robot.

The semantic map is authored truth, not perception

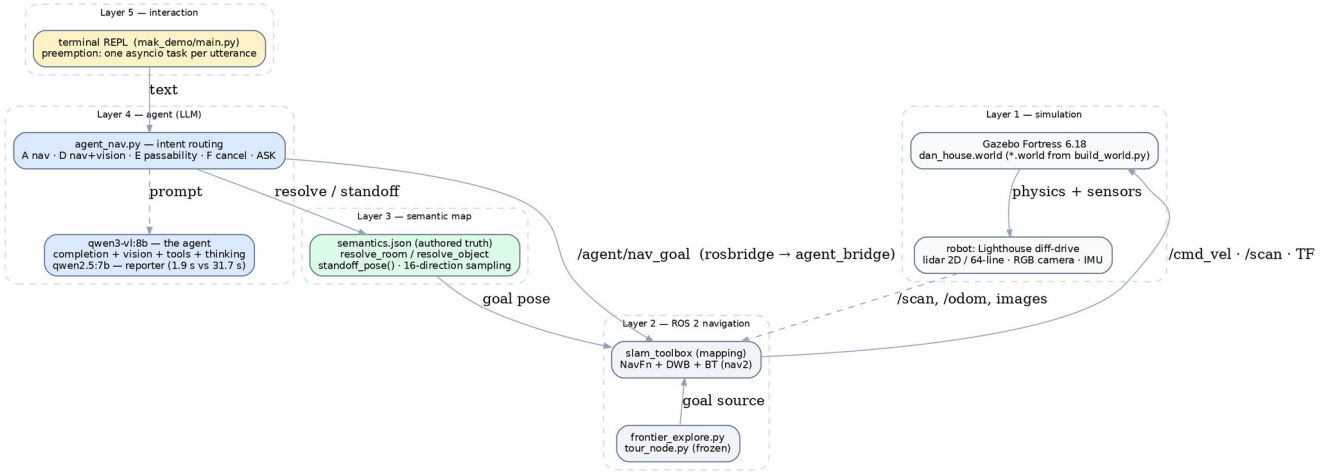


Fig. Layer 3 is a database: 8 rooms, 61 objects, 33 classes — generated with the world, read-only at runtime.

- Monocular camera: no depth, no scale → no map write-back
- Millimetre coordinates, because it comes from the same table as physics
- Standoff: 16 directions, clearance = inflation radius
- Observation pose: radius + 1.40 m, camera 0.70 m
- At 0.55 m the VLM calls a fridge "a door" — geometric fix, not a model fix

Changing its mind mid-drive

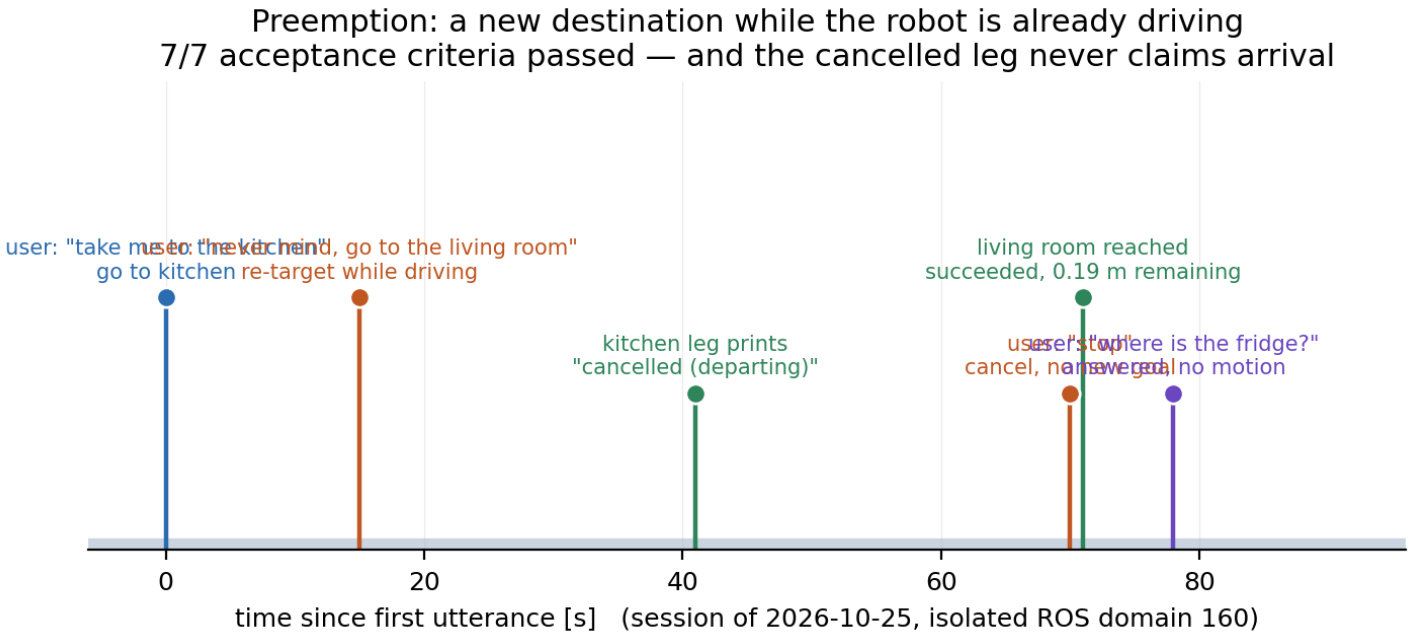


Fig. Isolated domain 160. The cancelled leg prints "cancelled (departing)" and never claims arrival. 7/7 criteria passed.

- Judgment order: new destination > question > cancel
- Three pieces: to_thread · one task per utterance · code-side table
- Heartbeat 0 vs 25 per 0.5 s proves why to_thread is required
- Four bugs only a live run could find (stale seq, stuck thread, merged input, cancel parsed as a place)
- Guardrails: source-level assertions in the offline suite

ACCEPTANCE

Eleven measurements, not eleven opinions

Gate	What it proves	Result (2026-10-26)
G1 / G1b	link + localisation	24 pose frames · 0.00 m / 0.0°
G2	hard-coded goal	succeeded 13.3 s · 0.19 m · 0 recoveries
G3	object → standoff → Nav2	succeeded 63.0 s · 0.15 m off plan
G4	mode D vision	1.79 m · VLM 13.1 s · 4/4 checks
G5 / G6	passability / question	laser 1.12 m + vision agree · 41.1 s
G7 / G8	refusal / compound	0.0 s · two segments succeeded
offline	11 checks + 6 guard cases	11/11 · 6/6

A cancelled leg may never report arrival — that single sentence summarises half of the preemption work.

HONEST GAPS

What this project deliberately does not do

Gap	Status	Why it matters
loc:=amcl	never re-verified	every number here assumes truth poses
EKF fusion drift	7.1 m / −37°, unsolved	blocks wheel-odometry-only running
3D obstacles in costmap	not implemented	overhangs and table tops are invisible
semantic map from perception	not attempted	monocular camera has no depth
coverage > 13.5 m²	measured ceiling	frontier exploration plateaus
voice / wake word	not implemented	the REPL is the only input
real hardware	never run	this is a simulation project throughout

Each row is a decision with a reason, not an oversight.

OUTLOOK

From a house to a warehouse: 4–6 days, three real changes

Change	Nature	Work
scale 18×14 m → 60×40 m	re-measure	range, loop closure, costmap size, budget
3D obstacles	architectural	voxel layer from /scan/points; overhangs become visible
slot-address semantics	architectural	A-12-03-2 → pose; new resolver, reuse standoff
world + parameters	data	new generator, lexicon, route, maps
everything else	unchanged	~3,900 lines: robot, launch, agent, tools, gates

Recommendation: add a world module in this repo (world:= switch, new branch), not a fork. Postponed on 2026-10-26 by decision.

Three things worth carrying away

1. **Keep authority apart.**

   Code moves the robot and refuses what cannot exist; the model resolves ambiguity and talks.

2. **Measure, then decide — and write down the retraction.**

   Clearance was a superstition until 3×3 said so. One published conclusion here was withdrawn.

3. **Make failures loud at the edges.**

   A second publisher on /scan, a stale goal result, a cancelled leg claiming arrival — each is silent by nature, so each got a mechanism.